

**Source: claim
infinitely many primes**

**Assumption: finite list
 p_1 through p_n**

**Algorithm capsule: N
equals product of all
listed primes plus 1**

**Assertion: N not
divisible by any listed
prime**

**Inference: N is prime
or has a new prime
factor**

**Contradiction: list was
complete**

**Conclusion: no finite
list is exhaustive**